# Sahaj Bhadja

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## Summary

Software engineer with 2+ years building C++ and Python oriented systems, concurrent workflows, and backend services. Strong in data structures, algorithms, OS fundamentals, debugging, and reliable software delivery.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*
- Maintained SQL-backed services for 500+ users, tuning indexed queries, background workers, and validation flows across FastAPI workflows to improve dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant LLM job orchestration with explicit state transitions, failure isolation, and CI/CD-backed releases, processing 5,000+ monthly workflows while improving backend reliability and operational visibility under concurrency.
- Collaborated on data fixes, release workflows, and service integrations, translating stakeholder requests into maintainable backend changes while documenting debugging paths that made recurring production issues faster to isolate.
- Supported asynchronous processing across production workflows, tracing failures through service boundaries and validating release behavior so platform updates remained stable while concurrent jobs and integrations scaled over time.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*
- Engineered API and ETL integrations using Python, SQL, AWS Glue, and S3, improving response time by 30% while supporting scalable service integrations, data collection workflows, and maintainable backend components.
- Maintained 20+ automated tests and CI pipelines with Jenkins, PyTest, and Mockito, increasing release reliability by 68% and strengthening debugging discipline across validation checks, deployments, and multi-environment builds.
- Worked within a microservice architecture spanning multiple operating systems, supporting executable builds, nightly validation runs, and cross-environment troubleshooting for production-style workflows with strict reliability and uptime expectations.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*
- Built Python evaluation pipelines across 100,000+ records, generating structured metrics, logs, and visual summaries for ML experiments while improving reproducibility, performance analysis, and reasoning about model behavior.
- Documented prompt, evaluation, and behavioral review workflows for researchers, clarifying output patterns and edge cases so experiments became easier to reproduce, debug, and compare across rapid iterations.
- Benchmarked model outputs across repeated runs and evaluation settings, helping researchers compare behavioral differences systematically while strengthening experiment traceability, debugging depth, and confidence in rigorous performance-driven conclusions.

## Skills

**Languages**: C++, Python, Java, JavaScript, TypeScript, SQL, Bash
**Systems**: data structures, algorithms, operating systems, object-oriented design, multithreading, concurrency, Linux, debugging, performance tuning
**APIs & Services**: FastAPI, Flask, Node.js, Express, REST APIs, WebSockets, JSON, API integration, microservices, background workers
**Data, ML & Delivery**: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, ETL pipelines, query tuning, indexing, PyTorch, Transformers, AWS, Azure, Docker, Jenkins, Git, GitHub Actions, CI/CD, PyTest, Mockito, unit testing, Agile/Scrum

## Projects

**CPU Scheduling Simulator** Live | *C++, React.js, EJ-2*
- Implemented FCFS, SJF, SRT, RR, priority scheduling, and MLQ with C++ logic, interactive controls, and dynamic metric calculation across 1,000+ simulated process combinations and execution scenarios.

**CaptionCraft** Live | *Transformers, PyTorch, Flask*
- Benchmarked BLIP, GIT, ViT-GPT-2, and custom transformer models on 10K images and 100K captions, improving caption accuracy by 15% while building reproducible inference and evaluation research workflows.

**Linguistix** | *Python, AWS, Docker, CI/CD*
- Developed a containerized Python REST API on AWS with automated CI/CD, validation checks, and reproducible releases, achieving 87% classification accuracy while supporting reliable data-processing workflows in production-style environments.